Database Foundations

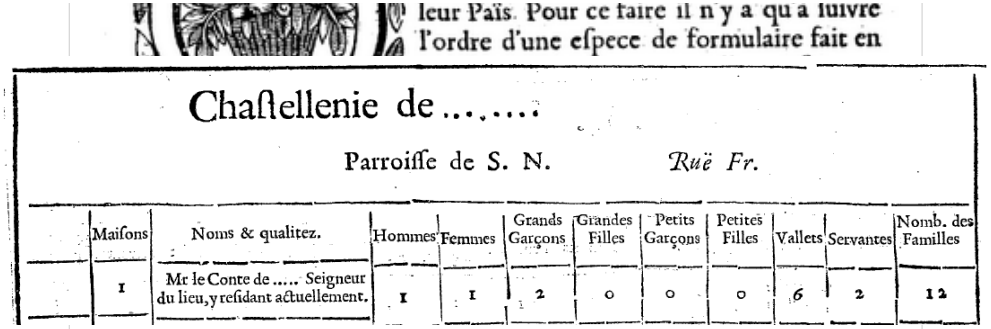
Option Presentation - M1 MOSIG

Silviu **Maniu, LIG,** Univ. Grenoble Alpes

Databases

Database: *organized collection* of data; *regular structure,* to capture a part of the *real world*

- Started with tabular data: records (employee, census, banking)
- Nowadays more varied data: medical databases, Web search logs, sensor logs, financial data (time-series)

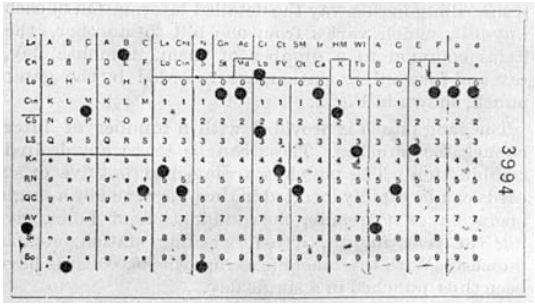


leur Païs. Pour ce faire il n'y a qu'a ſuivre l'ordre d'une eſpece de formulaire fait en

Chaſtellenie de .....

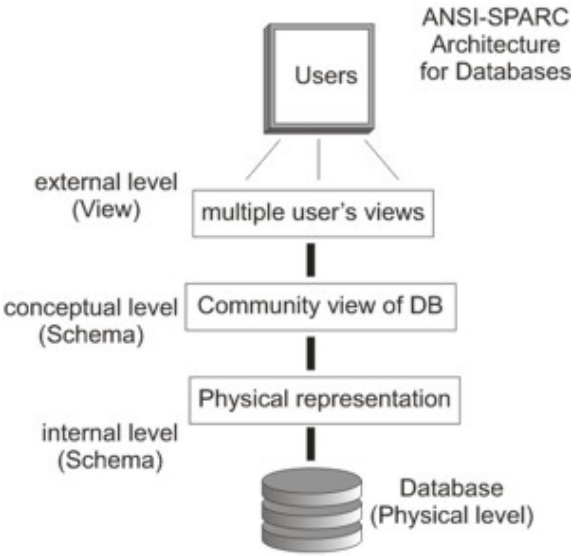
Parroiſſe de S. N. *Ruë Fr.*

	Maiſons	Noms & qualitez.	Hommes	Femmes	Grands Garçons	Grandes Filles	Petits Garçons	Petites Filles	Vallets	Servantes	Nomb. des Familles
	1	Mr le Conte de ..... Seigneur du lieu, y reſidant actuellement.	1	1	2	0	0	0	6	2	12





Database Management Systems



DBMS - dedicated *software* to manage databases: *data storage*, *data transactions*, *querying data*

Applications:

- Enterprise management: accounting, supply chain, staff, order
- Web applications: online retailers (Amazon), streaming providers, search engines
- Web backends: CMS, emails, certificates
- Scientific data: medicine, biology, etc.

ORACLE

IBM

Microsoft SQL Server

MySQL

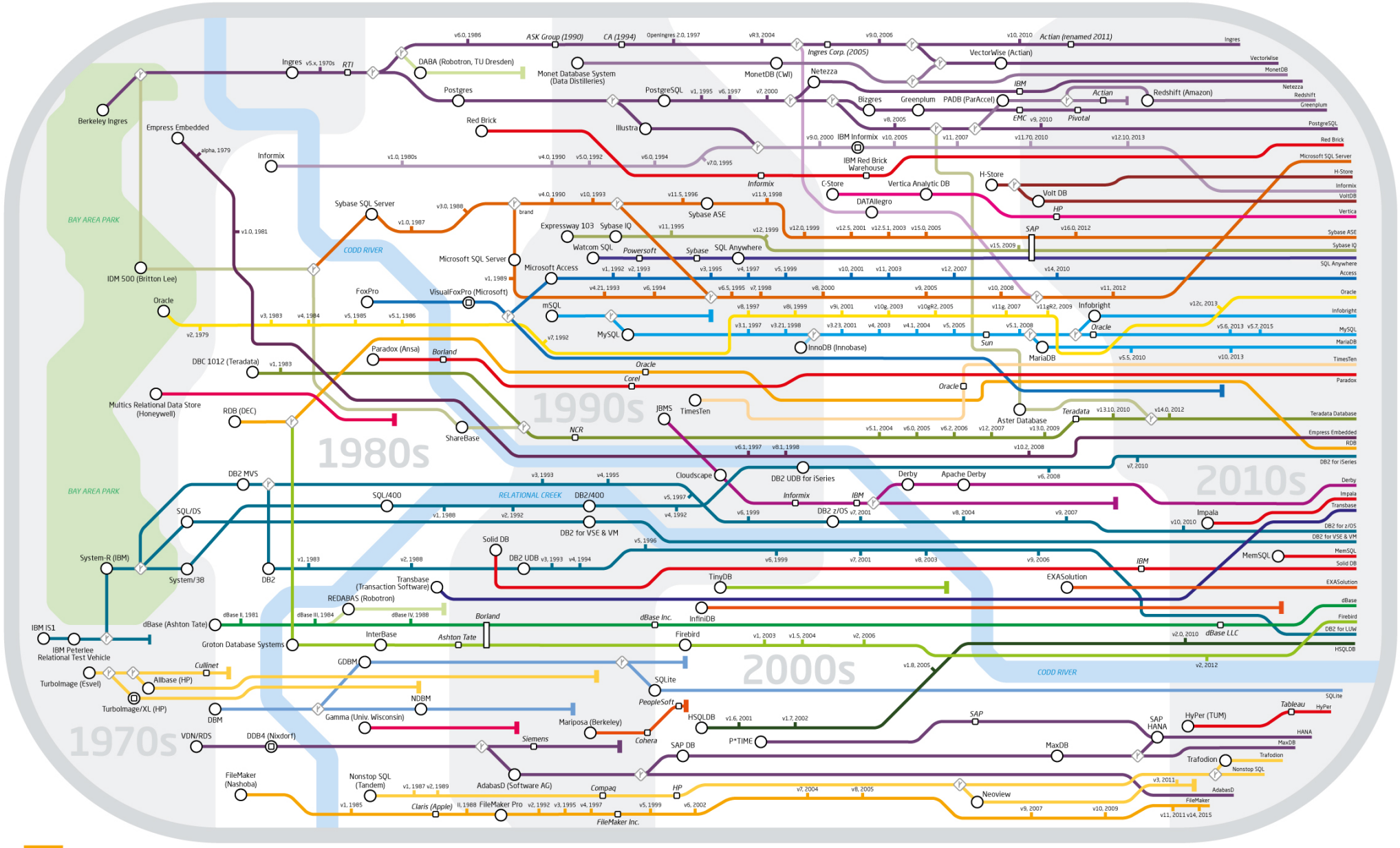
MariaDB

PostgreSQL

SQLite

mongoDB.

Genealogy of Relational Database Management Systems



Key to lines and symbols

DBMS name (Company) | *Acquisition* | v9, 2006 Versions | Discontinued | Branch (intellectual and/or code) | Crossing lines have no special semantics

Hasso Plattner Institut
Digital Engineering · Universität Potsdam

Felix Naumann, Jana Bauckmann, Claudia Exeler, Jan-Peer Rudolph, Fabian Tschirschnitz
Contact - Hasso Plattner Institut, University of Potsdam, felix.naumann@hpi.de
Design - Alexander Sandt Grafik-Design, Hamburg

Version 6.0 - October 2018
https://hpi.de/naumann/projects/rdbms-genealogy.html

Database Foundations

Objectives

1. To learn the **fundamental concepts** behind database models: relations, constraints, dependencies
2. To learn how to **design databases** for your application
3. To learn how to **query and program databases**
4. To learn how DBMS handle **transactions, locking...**

Database Foundations

Contents + Structure

- Applications of databases
- **Data models**: relational model, constraints, functional dependencies
- **Database design**: entity/relationship model, normalisation
- **Database programming**: relational algebra+SQL, procedural languages, triggers, views
- **Transactions**: ACID, isolation, 2-phase locking

“Cours-TD”: lecture + practical lab (exercises, implementation)

Evaluation: written exam + project (design of a DB application)